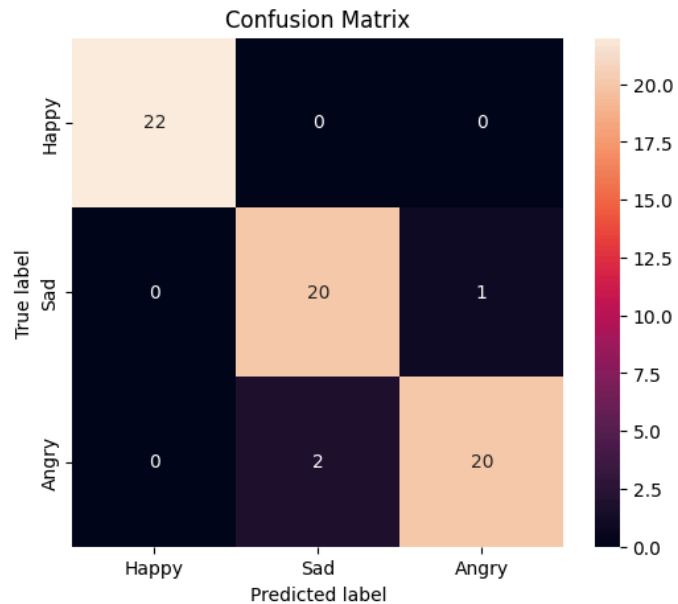


Face Emotion Recognition

Confusion Matrix



Model Performance Summary

- Accuracy: 0.9538
- Mean Cross-Validation Accuracy: 0.9075 (+/- 0.0751)
- Weighted Precision: 0.9545
- Weighted Recall: 0.9538
- Weighted F1-Score: 0.9538
- Misclassification Rate: 4.62%
- Misclassified Samples: 3/65

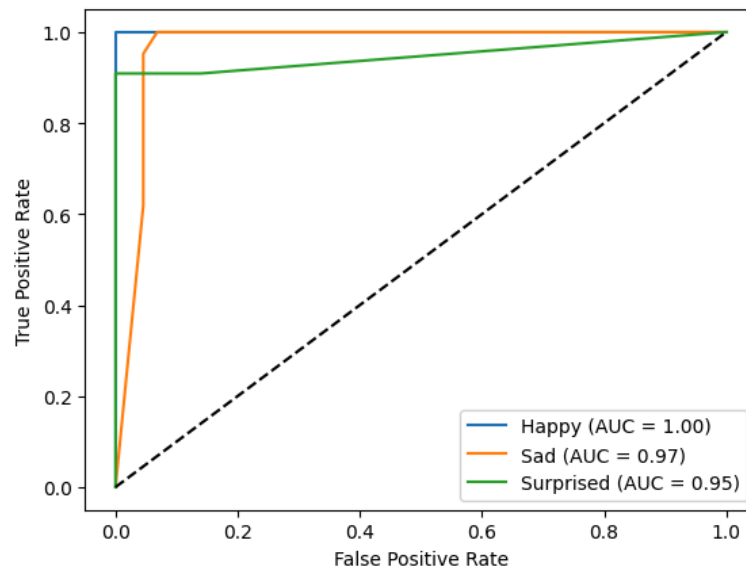
Sample Misclassifications:

- True: Sad Predicted: Surprised
- True: Surprised Predicted: Sad
- True: Surprised Predicted: Sad

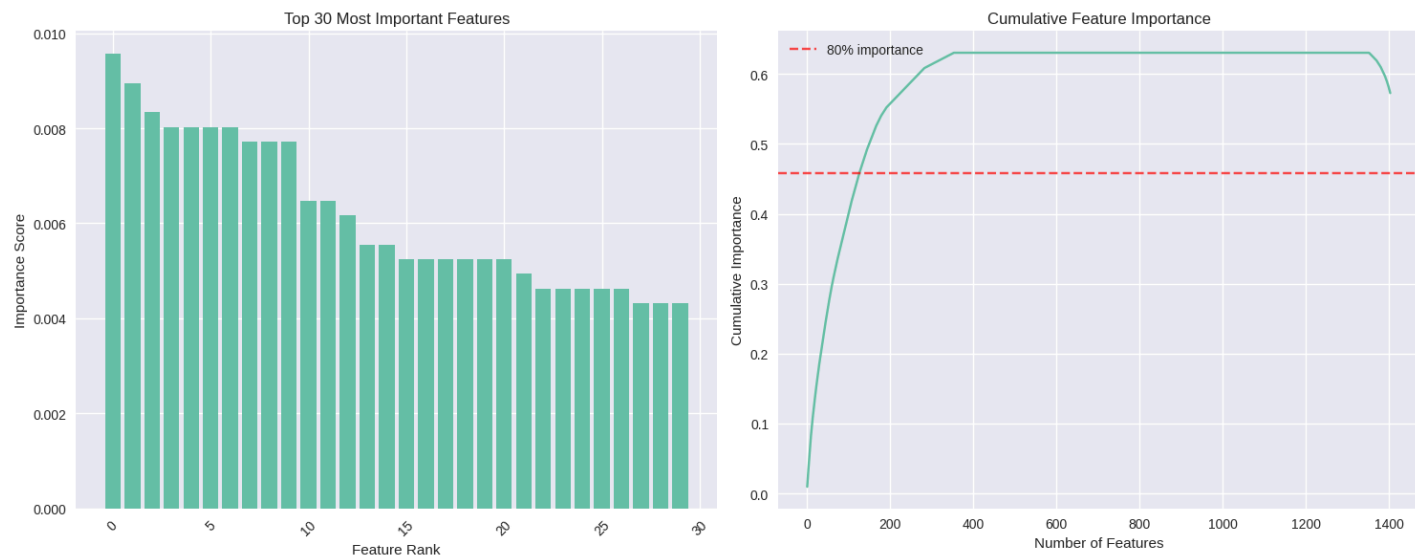
Class-wise Performance:

- Happy: Precision=1.000, Recall=1.000, F1=1.000
- Sad: Precision=0.909, Recall=0.952, F1=0.930
- Surprised: Precision=0.952, Recall=0.909, F1=0.930

ROC Curvers

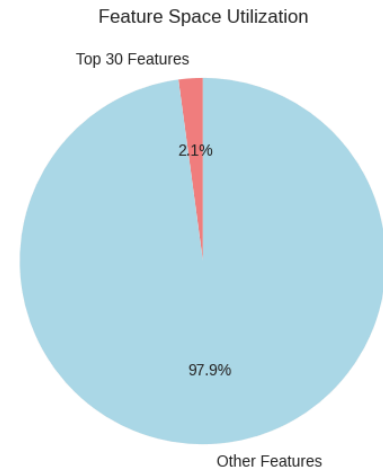
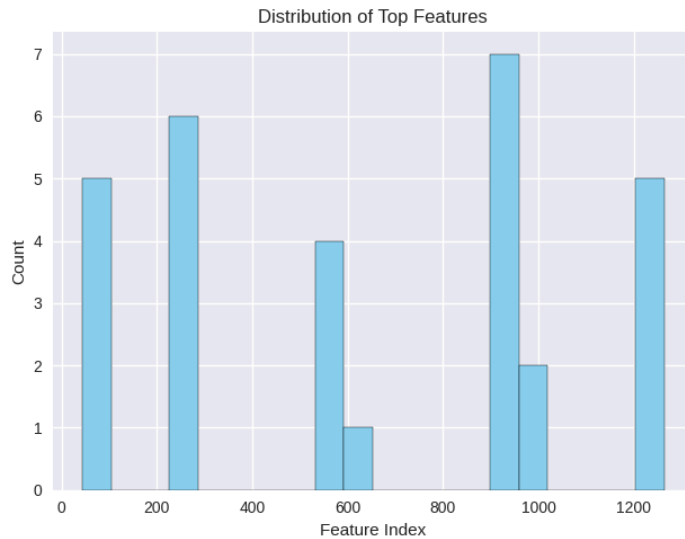


Feature Findings



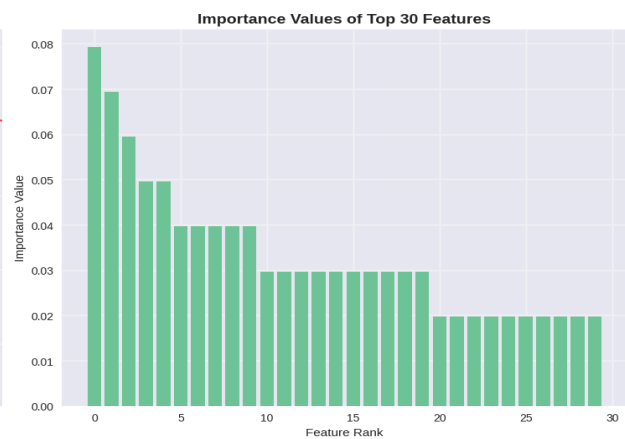
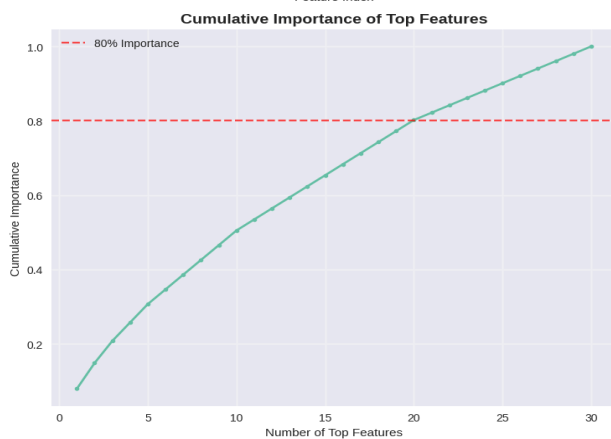
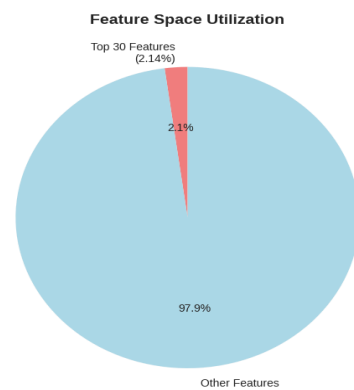
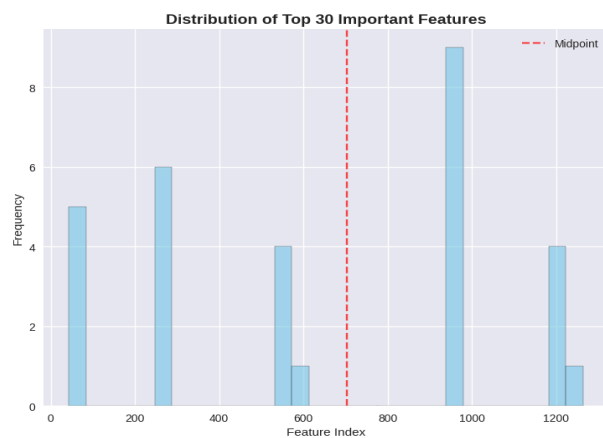
- Top 30 features: Account for significant predictive power
- 80% importance reached: 127 features (9.0% of total)
- 90% importance reached: 161 features (11.5% of total)
- Feature span: Index 43 to 1264 (wide distribution)
- Efficiency: Model uses only most discriminative features
- Robustness: No over-reliance on single features

Feature Distribution



- Feature span: Index 43 to 1264 (87% of range coverage)
- Utilization: 30 features (2.14% of total feature space)
- Distribution: Multi-modal with clusters at 200-400, 500-600, 900-1000, 1200-1264
- Coverage: Features distributed across 87% of possible range
- Efficiency: Optimal use of minimal feature set
- Pattern: Holistic rather than localized feature dependence

Comprehensive Feature Analysis



- Space efficiency: 30 features (2.14%) deliver majority predictive power
- Importance hierarchy: Top 5 features carry disproportionate importance
- Cumulative gain: 80% importance achieved rapidly with first features
- Distribution: Multi-modal pattern across feature space
- Value decay: Gradual importance decrease from top to bottom features
- Optimization: Ideal balance between complexity and performance

Conclusion

- 95.38% overall accuracy on test data
- Perfect classification of Happy expressions (100% precision/recall)
- Excellent generalization with 90.75% mean cross-validation accuracy
- Efficient feature utilization using only 2.14% of available features
- real-world deployment